NUMERICAL VERIFICATION OF STABILITY AND LOCAL POLYAK–ŁOJASIEWICZ INEQUALITIES FOR ENTROPIC MEAN-FIELD NEURAL ODES

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Abstract. Daudin and Delarue (arXiv:2507.08486) proved that the entropic mean-field control problem modeling deep residual networks with a continuum of layers admits, for an open and dense set of initial data, a unique *stable* minimizer $\boldsymbol{\nu}^*$, and that a local Polyak–Łojasiewicz (PL) inequality $\mathcal{I}(\boldsymbol{\nu}) \geq c\,(J(\boldsymbol{\nu}) - J(\boldsymbol{\nu}^*))$ holds around it, where $\mathcal{I}$ is the Fisher-type dissipation of the Wasserstein gradient descent, $\frac{d}{ds}J = -\mathcal{I}$. The exponential convergence of the descent is left there as a conjecture, the constants degenerate in an unquantified way as the entropic intensity $\varepsilon \to 0^+$, and no numerical realization of the results is available. We provide one. We build a solver for the explicit Gibbs representation of $\boldsymbol{\nu}^*$ (damped fixed-point iteration on the forward–backward optimality system, with the backward transport equation solved exactly along characteristics through the Jacobian cocycle), a discretization of $\mathcal{I}$ that vanishes exactly at the discrete Gibbs fixed point, a Scharfetter–Gummel scheme for the Wasserstein gradient flow whose equilibrium is exactly $\boldsymbol{\nu}^*$, and a matrix-free evaluation of the smallest Hessian eigenvalue. On a classification and a regression benchmark ($d_1 = 2$, five-dimensional parameter space) we test both results of Daudin–Delarue as stated. *Stability and uniqueness:* the fixed point is reached from initializations far from it, the projected Hessian is positive definite along the whole ε-schedule, and the Jacobi condition holds: the problem restarted from an intermediate layer recovers the restriction of the optimum and is itself stable. [pending: genericity across perturbed data.] *PL inequality:* every sampled perturbation satisfies it, the gradient descent converges exponentially with a rate that does not depend on the initial direction, and three independent estimators show that the PL constant degenerates essentially linearly in ε ($c(\varepsilon) \propto \varepsilon^{1.06}$, $\hat{c}(\varepsilon) \propto \varepsilon^{1.20}$ on the classification task, $c(\varepsilon) \propto \varepsilon^{0.99}$ on the regression task), with a super-linear steepening at the smallest ε only on the classification task, where the optimal control concentrates. We also find that the loss of convergence of the fixed-point iteration at small ε is a period-doubling of the damped map — not a loss of stability of $\boldsymbol{\nu}^*$ — and that the gradient flow recovers the exact fixed point there.

1. Introduction

Residual networks with a continuum of layers can be modeled as a mean-field optimal control problem [4, 6, 5]: one minimizes, over flows of probability measures $\boldsymbol{\nu} = (\nu_t)_{t\in[t_0,T]}$ on a parameter space $A = \mathbb{R}^{d'}$,

$$J\big((t_0,\gamma_0),\boldsymbol{\nu}\big) := \int_{\mathbb{R}^{d_1}\times\mathbb{R}^{d_2}} L(x,y)\,d\gamma_T(x,y) + \varepsilon\int_{t_0}^{T} \mathcal{E}\big(\nu_t \,\big|\, \nu^\infty\big)\,dt, \tag{1.1}$$

where the joint law γ_t of features and labels obeys the continuity equation driven by the mean-field vector field $\bar{b}(x,\nu_t) = \int_A b(x,a)\,d\nu_t(a)$, the prior $\nu^\infty \propto e^{-\ell}$ is of Gibbs type, and $\mathcal{E}(\cdot|\cdot)$ denotes relative entropy. The time t plays the role of the layer index; ν_t is the distribution of the neurons of layer t, in the spirit of the mean-field description of two-layer networks [8, 2, 10]. Unlike the one-layer case, the composition of flows makes the problem non-convex, even with the entropic term.

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In this framework Daudin and Delarue [3] proved two structural results. For an open and dense set $\mathcal{O}$ of initial conditions, the problem has a unique *stable* minimizer $\boldsymbol{\nu}^*$ — the second variation of J is non-degenerate there (Meta-Theorem 1.3 of [3]). Near $\boldsymbol{\nu}^*$ a local Polyak–Łojasiewicz inequality holds (Meta-Theorem 1.4):

$$\mathcal{I}\big((t_0,\gamma_0),\boldsymbol{\nu}\big) \;\geq\; c\,\Big(J\big((t_0,\gamma_0),\boldsymbol{\nu}\big) - J\big((t_0,\gamma_0),\boldsymbol{\nu}^*\big)\Big), \tag{1.2}$$

where $\mathcal{I}$ is a Fisher-information-type functional satisfying $\frac{d}{ds}J(\boldsymbol{\nu}^s) = -\mathcal{I}(\boldsymbol{\nu}^s)$ along the Wasserstein gradient flow $(\boldsymbol{\nu}^s)_{s\geq 0}$ in the descent time s.

Several points are left open in [3] and motivate this work. First, the exponential convergence of the gradient descent initialized near $\boldsymbol{\nu}^*$ is stated as a "reasonable conjecture" whose argument "would however deserve to be expanded". Second, all constants — and the set $\mathcal{O}$ itself — depend on ε, non-uniformly as $\varepsilon \to 0^+$, a limit the authors single out as a perspective of research. Third, no numerical realization of either meta-theorem is available: the optimality conditions form a coupled forward–backward system whose solution is nontrivial even in low dimension.

This paper is a numerical companion to [3]. We do not reproduce or extend its proofs; we compute the objects its theorems speak about and test the statements as they are formulated.

Contributions. (i) A solver for the explicit Gibbs representation of $\boldsymbol{\nu}^*$ (Theorem 2.7 of [3]), with the backward equation solved exactly along characteristics; we identify its breakdown at small ε as a period-2 cycle of the damped map, and show that the gradient flow recovers the exact fixed point there (Section 3). (ii) Discretizations of $\mathcal{I}$ and of the Wasserstein gradient flow for which the discrete $\boldsymbol{\nu}^*$ is an exact zero and an exact equilibrium, respectively. (iii) A picture of the computed optimum and of its concentration as ε decreases (Section 4). (iv) Tests of Meta-Theorem 1.3: independence of the fixed point from the initialization, positive definiteness of the Hessian along the schedule, the Jacobi condition, [pending: and persistence across perturbed data] (Section 5). (v) Tests of Meta-Theorem 1.4 and of the conjecture: three independent estimators of the PL constant and of its degeneration as $\varepsilon \to 0^+$ (Section 6). The implementation is open source, with machine-precision correctness tests.

Our headline findings on the classification benchmark: the PL inequality holds at every sampled perturbation around $\boldsymbol{\nu}^*$; the gradient descent converges exponentially with a rate that does not depend on the initial direction; and the PL constant follows $c(\varepsilon) \propto \varepsilon^{1.06}$, $\hat{c}(\varepsilon) \propto \varepsilon^{1.20}$ — an essentially linear degeneration in the entropic intensity, which steepens super-linearly at the smallest ε. On the regression benchmark, where the optimal control barely concentrates, the degeneration is linear with no steepening: $c(\varepsilon) \propto \varepsilon^{0.99}$ down to $\varepsilon = 0.02$ and $\lambda_{\min}(\varepsilon) \propto \varepsilon^{0.90}$.

Section 2 recalls the framework, Section 3 describes the numerical method, Section 4 presents the benchmarks and the computed optimum, Sections 5 and 6 test the two meta-theorems, and Section 7 discusses implications and limitations.

2. The entropic mean-field control problem

We recall the framework of [3] only to the extent needed to define the computed objects; no proofs are reproduced.

2.1. Setting.

The features-labels law $\gamma_t \in \mathcal{P}(\mathbb{R}^{d_1} \times \mathbb{R}^{d_2})$ evolves by

$$\partial_t \gamma_t + \operatorname{div}_x\Big(\int_A b(\cdot,a)\,d\nu_t(a)\,\gamma_t\Big) = 0, \qquad \gamma_{t_0} = \gamma_0, \tag{2.1}$$

i.e. γ_t is the law of (X_t, Y_0) with $\dot{X}_t = \bar{b}(X_t,\nu_t)$: features move along the layers while labels stay fixed. The control ν_t does not depend on x — all features at layer t see the same distribution of

neurons — which distinguishes the problem from standard mean-field control. Throughout we use the prototypical data of Example 1.1 in [3]:

$$b(x,a) = \tanh(a_1 \cdot x + a_2)\, a_0, \qquad a = (a_0, a_1, a_2) \in A = \mathbb{R}^{d'}, \quad d' = 2d_1 + 1, \tag{2.2}$$

with quartic prior potential $\ell(a) = c_1|a|^4 + c_2|a|^2$ ($c_1 = 0.05$, $c_2 = 0.5$). These choices satisfy the regularity assumption and the discriminating property of [3]; the latter is a formalization of the universal approximation property of the activation. (ReLU activations and a purely quadratic prior fall outside these assumptions.)

2.2. **First-order conditions and Gibbs representation.** Any minimizer satisfies (Theorem 2.7 of [3])

$$\nu_t^*(a) \;\propto\; \exp\Big(-\ell(a) - \frac{1}{\varepsilon}\,\Psi_t^*(a)\Big), \qquad \Psi_t(a) := \int_{\mathbb{R}^{d_1}\times\mathbb{R}^{d_2}} b(x,a)\cdot\nabla_x u_t(x,y)\, d\gamma_t(x,y), \tag{2.3}$$

where $(\boldsymbol{\gamma}^*, \boldsymbol{u}^*)$ solves the forward–backward system consisting of (2.1) coupled with the backward transport equation $-\partial_t u_t - \bar{b}(x,\nu_t)\cdot\nabla_x u_t = 0$, $u_T = L$. The backward equation has no Hamiltonian term: u is constant along characteristics, which is the key structural fact exploited by our solver (Appendix A). Formula (2.3) is a *necessary* condition: fixed points of the Gibbs map are critical points, and optimality has to be checked separately (Section 5).

2.3. **Stability, uniqueness and the Jacobi condition.** The meta-theorems are, in the authors' own terminology, *informal summaries*; every quantity we compute is anchored to their rigorous counterparts. Meta-Theorem 1.3 corresponds to [3, Theorem 4.1]. A minimizer is *stable* if the second variation of J at $\boldsymbol{\nu}^*$ is positive in every admissible direction, equivalently if the linearized optimality system has only the trivial solution. Theorem 4.1 contains two statements:

(a) (*Jacobi condition.*) For *any* initial condition (t_0, γ_0), any minimizer $\boldsymbol{\nu}^*$ and any $t_1 \in (t_0, T]$, the restriction $(\nu_t^*)_{t\in[t_1,T]}$ is the unique minimizer of the problem started from $(t_1, \gamma_{t_1}^*)$, and it is stable: $(t_1, \gamma_{t_1}^*) \in \mathcal{O}$.

(b) (*Genericity.*) The set $\mathcal{O}$ of initial conditions with a unique stable minimizer is open and dense in $[0,T]\times\mathcal{P}_3(\mathbb{R}^{d_1}\times\mathbb{R}^{d_2})$.

Numerically, (b) suggests that stability should persist under small perturbations of the data, and (a) can be tested directly by restarting the problem at intermediate layers (Section 5).

2.4. **The PL inequality.** Meta-Theorem 1.4 corresponds to [3, Theorem 5.1]: for every compact $\mathcal{K} \subset \mathcal{O}$ there exist $r, c > 0$ such that

$$\int_{t_0}^{T} \mathcal{E}\big(\nu_t \,\big|\, \nu_t^*\big)\, dt \;\le\; r^2 \quad\Longrightarrow\quad \mathcal{I}\big((t_0,\gamma_0),\boldsymbol{\nu}\big) \;\ge\; c\,\Big(J\big((t_0,\gamma_0),\boldsymbol{\nu}\big) - J\big((t_0,\gamma_0),\boldsymbol{\nu}^*\big)\Big), \tag{2.4}$$

i.e. the neighborhood is measured by the *integrated relative entropy* $\lambda^2 := \int \mathcal{E}(\nu_t|\nu_t^*)\, dt$ — the quantity we record for every perturbation in E1. The functional $\mathcal{I}$ is defined explicitly in [3, eq. (5.1)]:

$$\mathcal{I}(\boldsymbol{\nu}) = \int_{t_0}^{T}\int_A \Big|\, \varepsilon\nabla_a \log\nu_t(a) + \varepsilon\nabla_a\ell(a) + \nabla_a\Psi_t(a)\Big|^2\, d\nu_t(a)\, dt, \tag{2.5}$$

with Ψ_t evaluated along the forward–backward system driven by the *current* control $\boldsymbol{\nu}$ — this is, term by term, the expression our $\mathcal{I}_h$ discretizes. Daudin–Delarue also rewrite $\mathcal{I}$ as a Fisher information relative to the Gibbs measure of the current potential [3, eq. (5.3)],

$$\mathcal{I}(\boldsymbol{\nu}) = \varepsilon^2 \int_{t_0}^{T}\int_A \Big|\nabla_a \log\frac{\nu_t}{\Gamma_t[\boldsymbol{\nu}]}(a)\Big|^2\, d\nu_t(a)\, dt, \qquad \Gamma_t[\boldsymbol{\nu}] \propto e^{-\ell-\Psi_t/\varepsilon}, \tag{2.6}$$

and $\mathcal{I}(\boldsymbol{\nu}) = 0$ iff $\boldsymbol{\nu}$ solves the first-order conditions — in particular the Gibbs formula (2.3) cancels the integrand at $\boldsymbol{\nu}^*$.

Remark 2.1 (What is proved and what is conjectured)**.** Along the Wasserstein gradient flow, $\frac{d}{ds}J(\boldsymbol{\nu}^s) = -\mathcal{I}(\boldsymbol{\nu}^s)$; combined formally with (2.4), Grönwall's lemma would give $J(\boldsymbol{\nu}^s) - J^* \le (J(\boldsymbol{\nu}^0) - J^*)\, e^{-cs}$ near $\boldsymbol{\nu}^*$. Daudin and Delarue prove the inequality (2.4), not this decay: they call the exponential convergence a "reasonable conjecture", note that the dissipation identity "would however deserve to be expanded", and leave "the verification of these conjectures for future research". A rigorous proof would require well-posedness of the flow, the global validity of the dissipation identity, and that the trajectory remains in the neighborhood where the local inequality holds. Experiment E2 tests the conjectured decay directly; the trajectory-level statement suggested by the authors (convergence of $\boldsymbol{\nu}^s$ itself, cf. [9]) is tested in Section 6.2.

3. Numerical method

3.1. **Explicit solver: fixed-point iteration on the Gibbs map.** We discretize $A = [-R, R]^{d'}$ ($R = 2$) by a tensor grid with m_j nodes on axis j, all m_j *odd* so that $a = 0$ — the mode of the prior and a dynamically inert point ($b(x, 0) = 0$, $\partial_x b(x, 0) = 0$) — is a node. Two grids are used: a uniform *pilot* grid $m_j \equiv 5$ ($M = 3125$) and an *asymmetric* production grid $m = (7, 9, 5, 7, 5)$ ($M = 11\,025$) that better resolves the active axes (identified from the pilot marginals, Section 4). Time is discretized with $K = 10$ steps ($T = 1$), and γ_0 by the N data points. The grid makes the solver an analysis tool in low dimension: M grows like m^{2d_1+1}.

Given a control $\boldsymbol{\nu}$ on the grid, the forward pass integrates by RK4 the particles $X_i(t)$ together with the variational Jacobians $\Phi_i(t) = \partial X_i(t)/\partial X_i(0)$, interpolating ν_t linearly between slices. Since u is transported along characteristics,

$$\nabla_x u_t\big(X_i(t)\big) = \big[\Phi_i(T)\Phi_i(t)^{-1}\big]^\top \nabla_x L\big(X_i(T), y_i\big), \tag{3.1}$$

the *Jacobian cocycle*, at cost $O(Nd_1^3)$ per time slice and with no state-space mesh (Appendix A). The Gibbs map (2.3) is then evaluated with logsumexp normalization, and the fixed point is sought by damped iteration (Algorithm 1)

$$\nu^{(k+1)} = (1-\omega)\,\nu^{(k)} + \omega\,\nu_{\mathrm{Gibbs}}(\nu^{(k)}), \qquad \omega = 0.2, \tag{3.2}$$

stopped when $\mathrm{diffTV} := \max_k \mathrm{TV}(\nu_{t_k}^{(k+1)}, \nu_{t_k}^{(k)}) < \mathrm{tol} = 10^{-4}$. The cost of one iteration is $O(NMKd_1^2)$, dominated by the evaluation of b and $\partial_x b$ on all (particle, node) pairs.

Algorithm 1 Explicit solver (damped Gibbs iteration with continuation in ε)

Require: data $\{(x_i, y_i)\}_{i=1}^N$, grid $\{a_l\}_{l=1}^M$, schedule $\varepsilon_1 > \cdots > \varepsilon_n$, $T, K, \omega, \mathrm{tol}, \mathrm{max_iter}$
- $\log \nu_l^\infty \leftarrow -\ell(a_l) - \mathrm{logsumexp}_{l'}(-\ell(a_{l'}))$; $\quad \boldsymbol{\nu}^{\mathrm{warm}} \leftarrow (\nu^\infty, \dots, \nu^\infty)$
- **for** $j = 1, \dots, n$ **do**
  - $\boldsymbol{\nu} \leftarrow \boldsymbol{\nu}^{\mathrm{warm}}$
  - **for** $k = 0, \dots, \mathrm{max_iter} - 1$ **do**
    - $(X, \Phi) \leftarrow$ RK4 forward pass of particles and Jacobians under $\bar{b}(\cdot, \boldsymbol{\nu})$
    - $Z_{i,k} \leftarrow [\Phi_i(T)\Phi_i(t_k)^{-1}]^\top \nabla_x L(X_i(T), y_i)$ ▷ cocycle (3.1)
    - $\Psi_{k,l} \leftarrow \frac{1}{N}\sum_i b(X_i(t_k), a_l) \cdot Z_{i,k}$; $\quad \log \nu_{k,l}^{\mathrm{G}} \leftarrow \log \nu_l^\infty - \Psi_{k,l}/\varepsilon_j - \mathrm{logsumexp}_l(\cdot)$
    - $\boldsymbol{\nu}^{\mathrm{new}} \leftarrow (1-\omega)\boldsymbol{\nu} + \omega\,\boldsymbol{\nu}^{\mathrm{G}}$; $\quad \mathrm{diffTV} \leftarrow \max_k \mathrm{TV}(\nu_k^{\mathrm{new}}, \nu_k)$; $\quad \boldsymbol{\nu} \leftarrow \boldsymbol{\nu}^{\mathrm{new}}$
    - **if** $\mathrm{diffTV} < \mathrm{tol}$ **then break**
    - **end if**
  - **end for**
  - save $\boldsymbol{\nu}^{*,\varepsilon_j} \leftarrow \boldsymbol{\nu}$; $\quad \boldsymbol{\nu}^{\mathrm{warm}} \leftarrow \boldsymbol{\nu}$
- **end for**

Calibration of the stopping criterion. The residual diffTV measures the *damped* step; near the fixed point the true distance is $\mathrm{dist}(\nu^{(k)}, \nu^{\mathrm{fix}}) \approx \mathrm{diffTV}/\omega$ (effective tolerance $5 \cdot 10^{-4}$). This factor propagates quadratically into $\mathcal{I}_h(\boldsymbol{\nu}^*)$ (Section 3.2) and must be reported for the downstream experiments to be interpretable.

Continuation in ε and its breakdown. The factor $e^{-\Psi/\varepsilon}$ amplifies small changes of Ψ by $1/\varepsilon$, and the damped map loses contractivity at small ε. We follow the schedule $\varepsilon \in \{0.5, 0.2, 0.1, 0.05, 0.02\}$ with warm starts, and refine it between 0.05 and 0.02 with the intermediate values $\{0.04, 0.03, 0.025\}$ (Table 1). At $\varepsilon \geq 0.05$ the iteration converges in 18–39 iterations to $\mathrm{diffTV}/\omega \approx 4 \cdot 10^{-4}$ on both benchmarks. Below, the breakdown point depends on the problem and not on the continuation step: *moons* still converges at $\varepsilon = 0.04$ (slowly, 42 iterations) but fails from $\varepsilon = 0.03$ on, whereas *housing* converges down to $\varepsilon = 0.025$ and fails at $\varepsilon = 0.02$; the fine schedule does not change the $\varepsilon = 0.02$ iterate of the coarse one (TV difference $2 \cdot 10^{-10}$ for *moons*).

The nature of the failure is revealed by continuing the *moons* iteration at $\varepsilon = 0.03$ for 120 further steps: the cost alternates *exactly* between $J = 0.38742$ and $J = 0.39038$ with constant $\mathrm{diffTV} = 8.03 \cdot 10^{-2}$. The iteration is trapped in a *period-2 cycle*. For a damped map $\nu \mapsto (1-\omega)\nu + \omega G(\nu)$, a period-doubling occurs when an eigenvalue κ of the linearized Gibbs map DG crosses $1 - 2/\omega = -9$: the undamped update over-reacts with the wrong sign and the damping no longer compensates. This is a failure of the *algorithm*, not of the minimizer: the fixed point exists and is stable, and the gradient flow reaches it (Section 6.2(c)). Smaller damping, $\omega < 2/(1 - \kappa_{\min})$, would restore local convergence at the price of slower iterations; we instead recommend the Picard-then-flow pipeline of Section 6.2.

	moons					*housing*				
ε	iters	conv	J	$\frac{\mathrm{diffTV}}{\omega}$	acc	iters	conv	J	$\frac{\mathrm{diffTV}}{\omega}$	R^2_{tr}
0.5	25	yes	0.834	$4.2 \cdot 10^{-4}$	0.698	19	yes	0.273	$4.6 \cdot 10^{-4}$	0.472
0.2	26	yes	0.722	$4.1 \cdot 10^{-4}$	0.725	18	yes	0.267	$4.7 \cdot 10^{-4}$	0.494
0.1	27	yes	0.602	$4.8 \cdot 10^{-4}$	0.777	19	yes	0.259	$4.4 \cdot 10^{-4}$	0.517
0.05	39	yes	0.475	$4.4 \cdot 10^{-4}$	0.818	22	yes	0.249	$4.2 \cdot 10^{-4}$	0.540
0.04†	42	yes	0.436	$4.9 \cdot 10^{-4}$	0.840	18	yes	0.245	$4.4 \cdot 10^{-4}$	0.547
0.03†	200	no	0.390	0.401	0.870	20	yes	0.241	$4.5 \cdot 10^{-4}$	0.555
0.025†	200	no	0.368	0.514	0.915	19	yes	0.238	$4.4 \cdot 10^{-4}$	0.560
0.02	200	no	0.342	0.550	0.935	200	no	0.320	0.555	0.572

TABLE 1. ε-continuation on the pilot grid ($M = 3125$, $\approx 9\,\mathrm{s}$ per iteration): iterations, convergence, cost, calibrated distance to the fixed point, and task metric (training accuracy / R^2). †Fine schedule, warm-started from $\varepsilon = 0.05$. Non-converged rows report the last iterate (for *moons* at $\varepsilon = 0.03$, a point of a period-2 cycle). Symmetry error 0.0 in every row; the asymmetric grid ($M = 11025$) reproduces these values closely.

3.2. **Discretization of the Fisher functional.** On the grid, node masses $\nu_{k,l}$ stand for the probabilities of cells of constant volume, so $\nabla_a \log(\text{density})$ and $\nabla_a \log(\text{mass})$ coincide and we differentiate $\log \nu_{k,l}$ directly. With trapezoidal weights w_k in time,

$$\mathcal{I}_h(\boldsymbol{\nu}) = \sum_{k=0}^{K} w_k \sum_{l=1}^{M} \big|\, G_{k,l} \,\big|^2 \nu_{k,l}, \qquad G_{k,l} = D_a\Big[\varepsilon \log \nu_k + \varepsilon\, \ell + \Psi_k\Big](a_l), \tag{3.3}$$

where D_a is the per-axis finite-difference gradient (centered in the interior, one-sided at $\partial[-R, R]^{d'}$) and Ψ_k is assembled from the forward pass, reusing the cocycle vectors (3.1). We differentiate the log masses directly (logsumexp representation, $\log \nu$ floored at -40 on numerically empty nodes).

Discrete compatibility. Applying the *same* operator D_a to all three terms in (3.3) is essential: the discrete Gibbs fixed point satisfies $\varepsilon \log \nu_k^* = -\varepsilon\ell - \Psi_k + \mathrm{const}_k$ exactly, so $G \equiv 0$ at $\boldsymbol{\nu}^*$ and $\mathcal{I}_h(\boldsymbol{\nu}_\varepsilon^*) \sim (\mathrm{diffTV}_{\mathrm{final}}/\omega)^2$. Equivalently, since $\varepsilon \log \nu + \varepsilon\ell + \Psi = \varepsilon \log(\nu/\Gamma[\boldsymbol{\nu}]) + \mathrm{const}$, the compatible assembly is the exact discretization of the Fisher-information rewrite (2.6), and the discrete functional inherits the property $\mathcal{I}_h = 0 \Leftrightarrow \nu = \Gamma_h[\boldsymbol{\nu}]$ (discrete Gibbs) by construction. The mixed assembly (analytic $\nabla\ell$, $\nabla_a\Psi$, finite-difference $\nabla_a \log \nu$) breaks this cancellation and leaves an $O(h^4)$-scale floor in $\mathcal{I}_h(\boldsymbol{\nu}^*)$. Table 2 shows the calibration on the pilot grid: $\mathcal{I}_h(\boldsymbol{\nu}_\varepsilon^*)$ tracks $(\mathrm{diffTV}_{\mathrm{final}}/\omega)^2$ within roughly one order of magnitude while the two quantities themselves range over five orders — tiny wherever the iteration converged, and large ($\sim 10^{-2}$) exactly at the non-converged endpoint $\varepsilon = 0.02$. The same table is a direct check of the Gibbs formula (2.3) at the computed controls.

	moons			*housing*		
ε	$\mathcal{I}_h(\boldsymbol{\nu}^*)$	$(\mathrm{diffTV}/\omega)^2$	ratio	$\mathcal{I}_h(\boldsymbol{\nu}^*)$	$(\mathrm{diffTV}/\omega)^2$	ratio
0.5	$3.4{\cdot}10^{-6}$	$1.8{\cdot}10^{-7}$	19.0	$1.5{\cdot}10^{-6}$	$2.1{\cdot}10^{-7}$	6.8
0.2	$2.5{\cdot}10^{-7}$	$1.7{\cdot}10^{-7}$	1.5	$2.3{\cdot}10^{-7}$	$2.2{\cdot}10^{-7}$	1.1
0.1	$5.1{\cdot}10^{-8}$	$2.3{\cdot}10^{-7}$	0.22	$5.4{\cdot}10^{-8}$	$1.9{\cdot}10^{-7}$	0.28
0.05	$4.1{\cdot}10^{-9}$	$1.9{\cdot}10^{-7}$	0.02	$1.1{\cdot}10^{-8}$	$1.7{\cdot}10^{-7}$	0.07
0.02	$9.1{\cdot}10^{-3}$	$3.0{\cdot}10^{-1}$	0.03	$4.6{\cdot}10^{-2}$	$3.1{\cdot}10^{-1}$	0.15

TABLE 2. Calibration gate on the pilot grid: the compatible assembly makes $\mathcal{I}_h(\boldsymbol{\nu}_\varepsilon^*)$ scale with the calibrated squared distance to the fixed point across the whole schedule.

3.3. **Gradient-flow scheme.** The Wasserstein gradient flow acts slice-wise in the descent time s,

$$\partial_s \nu_t^s = \mathrm{div}_a\big(\nu_t^s \, \nabla_a \Phi_t^{(s)}\big) + \varepsilon \Delta_a \nu_t^s, \qquad \Phi_t^{(s)} := \Psi_t^{(s)} + \varepsilon\,\ell, \tag{3.4}$$

with no-flux boundary conditions; the $K+1$ slices are coupled only through $\Psi^{(s)}$, which depends on the whole curve $\boldsymbol{\nu}^s$. We integrate with an explicit finite-volume scheme whose face flux is of Scharfetter–Gummel (exponential-fitting) type [11, 1]: across the face between neighboring cells L, R on axis j (spacing h_j),

$$F_{LR} = \frac{\varepsilon}{h_j^2}\Big[B(-z)\, m_L - B(z)\, m_R\Big], \qquad z = \frac{\Phi_L - \Phi_R}{\varepsilon}, \qquad B(x) = \frac{x}{e^x - 1}. \tag{3.5}$$

The scheme conserves mass exactly, preserves positivity under an explicit CFL bound (computed from the assembled outflow rates), and reduces to centered diffusion for $z \to 0$ and to upwind for $|z| \gg 1$. Its decisive property is that its zero-flux condition $F_{LR} = 0$ forces $m_R/m_L = e^{(\Phi_L - \Phi_R)/\varepsilon}$, the discrete Gibbs measure of the current potential: $\boldsymbol{\nu}^*$ is an *exact* equilibrium of the discrete flow, so the E2 decay curves carry no spurious floor. A plain upwind drift has an $O(h)$-diffused equilibrium $\neq \boldsymbol{\nu}^*$, which we observed to inject an $O(1)$ offset in $J(\boldsymbol{\nu}^s) - J^*$ on coarse grids.

The flow runs on the pilot grid, and Ψ is frozen for $m = 5$ steps in s between forward passes (identical J-decrements for $m \in \{1, 5, 10\}$ at $\varepsilon = 0.05$). The time step is set by the exact positivity (CFL) condition computed from the assembled outflow rates, and no-flux boundary conditions are imposed by omitting boundary faces, so that mass is conserved by telescoping.

The identity $dJ/ds = -\mathcal{I}$. As an integrated correctness test we verify $[J(\boldsymbol{\nu}^{s+\Delta s}) - J(\boldsymbol{\nu}^s)]/\Delta s \approx -\mathcal{I}_h(\boldsymbol{\nu}^s)$. The $O(\Delta s)$ error is negligible already at $\Delta s = 10^{-2}$, and the residual mismatch is the $O(h^2)$ grid error: at the tiled prior ($\varepsilon = 0.05$, *moons*) on the grids $m_j \equiv 3, 5, 7$ it is 0.206, 0.042 and 0.021, i.e. $\mathrm{rel}/h^2 \approx 0.05$ on every grid.

3.4. **Matrix-free Hessian.** With $(K+1)M \approx 3.4 \cdot 10^4$ unknowns the Hessian is never assembled. The Euclidean gradient of the discrete cost is $G_{k,l}(\boldsymbol{\nu}) = w_k[\Psi_l(t_k;\boldsymbol{\nu}) + \varepsilon(\log \nu_{k,l} - \log \nu_l^\infty)]$ modulo one constant per slice, removed by projecting onto zero-mean slices (the tangent space of the simplex constraints). Hessian-vector products split into an *exact* diagonal entropic part and a smooth finite-difference part:

$$(Hv)_{k,l} = \underbrace{\frac{w_k\,\varepsilon}{\nu^*_{k,l}}\,v_{k,l}}_{\text{analytic}} + \underbrace{\frac{G^\Psi(\boldsymbol{\nu}^* + hv) - G^\Psi(\boldsymbol{\nu}^* - hv)}{2h}}_{\text{2 forward passes}}, \qquad G^\Psi_{k,l}(\boldsymbol{\nu}) = w_k \Psi_l(t_k;\boldsymbol{\nu}), \tag{3.6}$$

with $h \sim 10^{-5}$. The split matters: differencing the full gradient would difference $\varepsilon \log \nu$ across numerically empty nodes, where any step crosses zero and clipping destroys both accuracy and the symmetry of H; $\Psi(\boldsymbol{\nu})$ instead contains no logarithms and is smooth in $\boldsymbol{\nu}$. The smallest eigenvalue $\lambda_{\min}$ of the projected operator is the discrete counterpart of the stability notion of [3]: $\lambda_{\min} > 0$ means that the second variation is positive in every tangent direction. Plain Lanczos on the projected operator stalls: the projection leaves a large null space (per-slice constants, off-support nodes) with eigenvalue $0 < \lambda_{\min}$, which the solver chases indefinitely. Penalizing the complement of the tangent subspace ($\sigma = 100$) and preconditioning LOBPCG with the exact entropic diagonal $\varepsilon w_k/\nu^*$ resolves it (validated against a dense eigendecomposition on a 3^5 test grid to $< 10^{-3}$ relative). Internal checks: Hessian symmetry to $4 \cdot 10^{-8}$ on random tangent pairs, and the quadratic model predicts the measured excess of a small multiplicative perturbation to 3% ($\langle H\eta, \eta\rangle/2 = 9.85 \cdot 10^{-6}$ vs $9.55 \cdot 10^{-6}$).

3.5. **Correctness tests.** Besides the calibration gate (Table 2) and the identity $dJ/ds = -\mathcal{I}_h$, the test suite verifies at machine precision the structural symmetry of Appendix B ($\nu_t^*(-a) = \nu_t^*(a)$, error $\sim 10^{-16}$), its preservation along the discrete flow, exact mass conservation and positivity of the scheme (3.5), the exact equilibrium property of the discrete Gibbs measure, the analytic $\nabla_a \Psi$ blocks against small-step finite differences at off-grid parameter points, the gradient G against directional derivatives of J, and $\det \Phi_i(t) > 0$ along every trajectory (so that the cocycle is well defined).

4. Benchmarks and the computed optimum

4.1. **Benchmarks.** Both benchmarks share $d_1 = 2$, $d' = 5$, $N = 400$, $T = 1$, $K = 10$, and the frozen seed of Appendix C. The features enter the flow as they are ($X_i(0) = x_i$, no augmentation): this keeps the parameter grid in $\mathbb{R}^5$.

(a) moons. `make_moons` (400 points, noise 0.12, standardized), two interleaved half-circles that no straight line separates. Labels are embedded as *anchors* on the first coordinate, $y \mapsto y_{\text{emb}} = (\mp 1, 0)$, with the quadratic loss of the prototypical setting of [3], $L(x,y) = \frac{1}{2}|x - y_{\text{emb}}|^2$, and prediction $\mathbf{1}\{X_1(T) > 0\}$. With symmetric anchors on one axis, the flow only has to push each class to its side of the hyperplane $x_1 = 0$, a one-dimensional transport task; this gives the best accuracy among the label embeddings we tried (0.935 at $\varepsilon = 0.02$).

(b) housing. California Housing restricted to the two features most correlated with the target on the training split (MedInc, $\rho = 0.73$; AveOccup, $\rho = -0.27$), standardized. The loss is scalar, $L(x,y) = \frac{1}{2}(x_1 - y)^2$, with prediction $X_1(T)$ and x_2 *free*. The alternative $\frac{1}{2}|x - (y,0)|^2$ would add a spurious penalty $\frac{1}{2}x_2^2$ that forces the flow to spend control on driving an auxiliary coordinate to zero; the scalar loss measures only the regression error ($R^2_{\text{tr}} = 0.572$ at $\varepsilon = 0.02$).

Both losses have unbounded gradients $\nabla_x L$, as in the quadratic setting of [3].

4.2. **What the optimal control looks like.** Before testing the theorems on $\boldsymbol{\nu}^*$ we show the object. Figure 1 displays the trajectories induced by the optimal mean field at $\varepsilon = 0.05$. On *moons* the flow pushes most of each class toward its anchor, to its side of $x_1 = 0$ (accuracy 0.82);

the entropic cost limits how far the particles travel, and the accuracy rises to 0.93 at $\varepsilon = 0.02$ (Section 6.2(c)). On *housing* the trajectories are short and nearly parallel to x_1: the field stretches the read-out coordinate along the target ($R^2_{\mathrm{tr}} = 0.54$) and leaves x_2 as a free auxiliary direction.

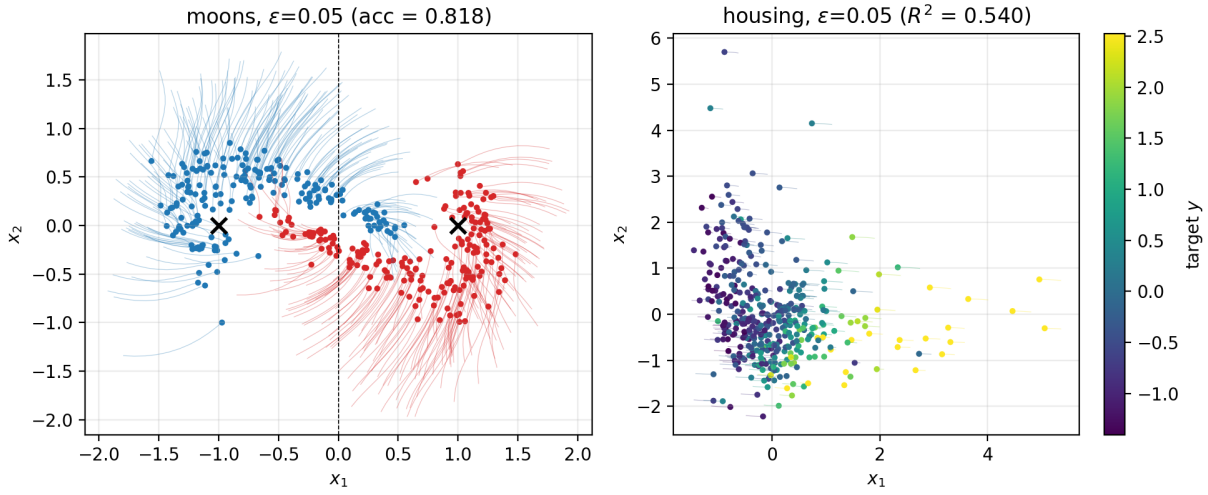


FIGURE 1. Trajectories $t \mapsto X_i(t)$ under the optimal mean field $\bar{b}(x, \nu_t^*)$ at $\varepsilon = 0.05$ (asymmetric grid). Left: *moons*, colored by class, filled dot = $X_i(T)$, crosses = anchors $(\pm 1, 0)$. Right: *housing*, colored by the target.

Figure 2 shows the one-dimensional marginals of ν_t^* against the prior. Two structural features are visible. First, the marginals are symmetric: the Gibbs formula makes ν_t^* exactly invariant under $a \mapsto -a$, so $\mathbb{E}_{\nu_t^*}[a] = 0$ (Appendix B); this reflects the mirror redundancy of weight space ($b(x, -a) = b(x, a)$), not a cancellation of the field, and the activity of the control shows in the even moments. Second, the control is sparse in the axes. On *moons* the output amplitudes $a_0^{(1)}, a_0^{(2)}$ and the input weight $a_1^{(2)}$ leave the prior, widening or becoming bimodal; the bias a_2 concentrates at zero, more sharply than the prior, and $a_1^{(1)}$ departs from the prior only in the first layers. On *housing* the activation is much weaker and concentrated on $a_0^{(1)}$ and $a_1^{(2)}$, consistent with a nearly linear regression. This diagnosis is what motivated the asymmetric grid of Section 3.1 (more nodes on the active axes).

Concentration of the optimal control as ε decreases. Ranking the grid nodes ("neurons") by their time-averaged mass $\bar{w}_l = \frac{1}{K+1} \sum_k \nu_{t_k}^*(a_l)$, Table 3 measures how much of the grid the optimal control actually uses: the effective support $N_{\mathrm{eff}} = \exp(H(\bar{w}))$, the number $n_{50\%}$ of nodes carrying half the mass, and the overlap of the ten heaviest nodes between the first and the last layer. The two benchmarks behave differently. On *moons* the mass concentrates as ε decreases (N_{eff} falls from 290 nodes at $\varepsilon = 0.5$ to 145 at $\varepsilon = 0.05$ and 81, 2.6% of the grid, at $\varepsilon = 0.02$) and the ranking becomes layer-dependent (only 4 of the top-10 nodes at $t = 0$ remain at $t = T$ for $\varepsilon = 0.05$): the optimal network uses different neurons at different depths. The dominant pairs are high-gain directional detectors: zero bias, input weights aligned with axes or diagonals, large output amplitudes. On *housing* the concentration is essentially insensitive to ε (N_{eff} stays between 284 and 295 down to $\varepsilon = 0.02$) and the ranking is largely stable in depth (7–9 of the top-10 nodes persist from $t = 0$ to $t = T$). This concentration of ν_ε^* is the mechanism invoked in Section 7.1 to interpret the degeneration of the PL constant.

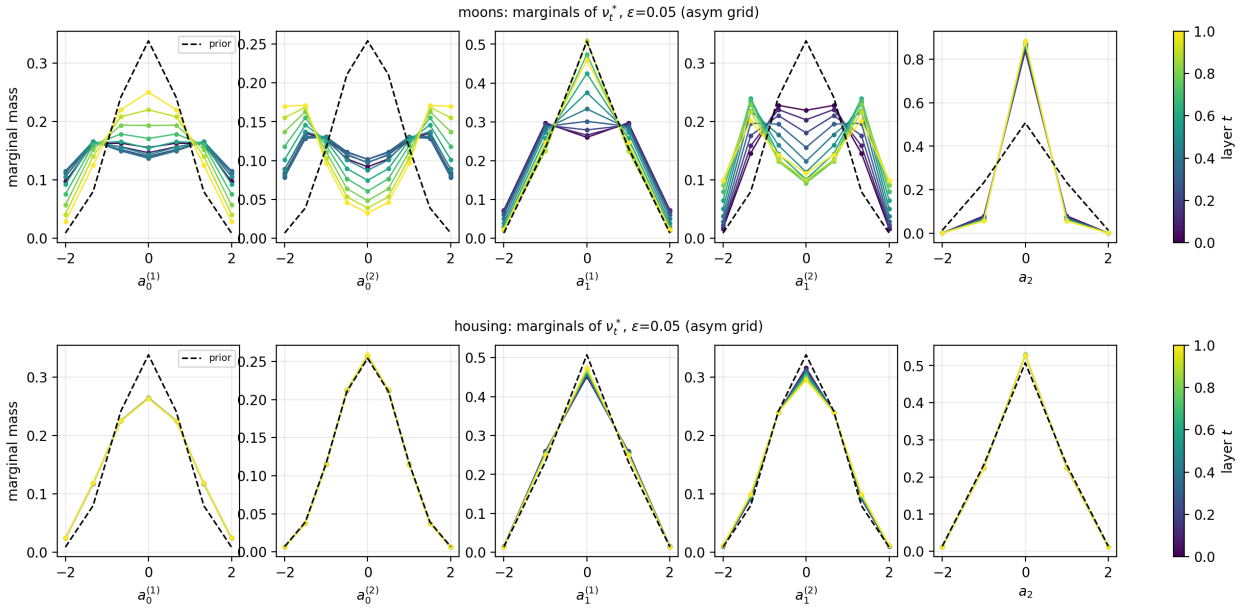


FIGURE 2. One-dimensional marginals of ν_t^* at $\varepsilon = 0.05$ (color = layer time t, dashed = prior ν^∞) on the asymmetric grid. Top: *moons*. Bottom: *housing*.

benchmark	ε	mass of top-1%	N_{eff}	$n_{50\%}$	top-10 overlap $t=0$ vs $t=T$
moons	0.5	0.33	290	60	7.5/10
moons	0.1	0.48	202	35	6.0/10
moons	0.05	0.55	145	26	4.0/10
moons	0.02†	0.69	81	17	4.0/10
housing	0.5	0.31	295	62	9.1/10
housing	0.1	0.32	293	61	7.5/10
housing	0.05	0.33	289	61	7.3/10
housing	0.02†	0.33	284	61	7.0/10

TABLE 3. Concentration of $\boldsymbol{\nu}_\varepsilon^*$ on the pilot grid ($M = 3125$). $N_{\text{eff}} = \exp(\text{Shannon entropy of } \bar{w})$. The $a \leftrightarrow -a$ symmetry makes node weights come in exactly equal pairs (and, at large ε, larger groups), so the top-10 overlap is the expected overlap when nodes tied at the cut-off are chosen at random. †Flow-refined endpoint (Section 6.2(c)).

5. Stability and uniqueness (Meta-Theorem 1.3)

5.1. **Uniqueness: independence from the initialization.** The solver starts from the prior and follows the continuation, which could in principle select one critical point among several. We remove both ingredients: we start the iteration from the *uniform* measure on the grid, $\nu_t^{(0)} \equiv 1/M$, far from the fixed point ($\mathrm{TV}(1/M, \nu^*) \approx 0.9$), and run it *directly* at each ε, without continuation. All six runs (Table 4) converge in 35–45 iterations to the same fixed point as the continuation, at distance of the order of the effective tolerance tol/ω, with exactly zero symmetry error. At these data and for $\varepsilon \geq 0.05$, the critical point found is therefore not an artifact of the initialization or of the continuation path. This is consistent with the uniqueness part of Meta-Theorem 1.3; the continuation is needed only to approach the regime $\varepsilon < 0.05$.

benchmark	ε	iters	conv	J_{unif} / J_{cont}	$\text{TV}(\nu_{\text{unif}}, \nu_{\text{cont}})$
moons	0.5	35	yes	0.83385 / 0.83356	$4.6 \cdot 10^{-4}$
moons	0.1	37	yes	0.60247 / 0.60242	$4.0 \cdot 10^{-4}$
moons	0.05	45	yes	0.47521 / 0.47520	$1.9 \cdot 10^{-4}$
housing	0.5	35	yes	0.27349 / 0.27321	$5.6 \cdot 10^{-4}$
housing	0.1	35	yes	0.25888 / 0.25882	$5.4 \cdot 10^{-4}$
housing	0.05	35	yes	0.24880 / 0.24877	$5.2 \cdot 10^{-4}$

TABLE 4. Uniform initialization without continuation (pilot grid). The TV distance (max over layers) to the continuation fixed point is at the level of the effective tolerance $\text{tol}/\omega = 5 \cdot 10^{-4}$.

5.2. **E3: strict stability along the schedule.** Table 6 reports $\lambda_{\min}(\varepsilon)$ of the projected Hessian at $\boldsymbol{\nu}_\varepsilon^*$ (Section 3.4; 103–308 Hessian-vector products per value). On *moons* it decreases from 1.23 at $\varepsilon = 0.5$ to $4.42 \cdot 10^{-2}$ at $\varepsilon = 0.05$ and stays strictly positive: the computed minimizers are stable in the sense of [3] across the schedule — a quantitative version of Meta-Theorem 1.3 at these data. Together with E1 (every sampled perturbation increases J, Section 6.1) this confirms that the fixed points are strict local minima and not saddles. The $\lambda_{\min}$ curve is not a clean power law: its local log-log slope steepens from 1.32 ($\varepsilon \in [0.2, 0.5]$) to 1.71 ($\varepsilon \in [0.05, 0.1]$), a super-linear degeneration discussed in Section 7.1. On *housing* $\lambda_{\min} = 1.06$, 0.236 and 0.134 at $\varepsilon = 0.5$, 0.1 and 0.05: strictly positive as well, and degenerating more slowly, with slope 0.90 (local slopes 0.81 and 0.93) and no steepening. [pending: *moons* and *housing* at the flow-refined $\varepsilon = 0.02$, and *housing* at $\varepsilon = 0.2$.]

5.3. **Genericity: stability under perturbations of the data.** Meta-Theorem 1.3 is a statement about *sets* of initial conditions, whereas the experiments above fix one γ_0 per benchmark. Since $\mathcal{O}$ is open, a stable minimizer should persist, with a nearby value of $\lambda_{\min}$, under small perturbations of γ_0; since it is dense, a randomly drawn data set should fall in $\mathcal{O}$. We probe this with families of data sets: *moons* with five sampling seeds and noise levels $\{0.08, 0.12, 0.16\}$, and *housing* with five training subsamples. For each data set we recompute the continuation on the pilot grid down to $\varepsilon = 0.05$, $\lambda_{\min}$ at $\varepsilon \in \{0.1, 0.05\}$, and a reduced E1 cloud (30 perturbations) giving $c(\varepsilon)$ (Table 5). [pending: Table 5: expected outcome to be confirmed — $\lambda_{\min} > 0$ for every data set, with small spread, and the same log-log slope of $c(\varepsilon)$.]

benchmark	data variant	$\lambda_{\min}(0.1)$	$\lambda_{\min}(0.05)$	$c(0.1)$	$c(0.05)$
moons	seeds 0–4, noise 0.12	pend.	pend.	pend.	pend.
moons	noise 0.08 / 0.16	pend.	pend.	pend.	pend.
housing	subsamples 0–4	pend.	pend.	pend.	pend.

TABLE 5. Genericity probe: range (min–max) over the data variants. [pending: not yet computed.]

5.4. **The Jacobi condition.** Statement (a) of Theorem 4.1 can be tested without any perturbation of the data. For a layer index $k_1 \in \{2, 4, 6, 8\}$ we restart the problem at $(t_{k_1}, \gamma^*_{t_{k_1}})$: the particles start at the optimal positions $X_i^*(t_{k_1})$ and the horizon is $[t_{k_1}, T]$. We solve it from the prior, without warm start, and compare the result with the restriction $(\nu_t^*)_{t \geq t_{k_1}}$ of the full optimum. The theorem predicts (i) that the two coincide (dynamic programming plus uniqueness) and (ii) that the restarted minimizer is stable, $\lambda_{\min}(t_{k_1}) > 0$. We run this at $\varepsilon = 0.05$ on *moons*. The trapezoidal weight of the first slice of the restarted problem differs from its weight in the full problem; the Gibbs fixed point does not depend on the time weights, so this affects only the Hessian, which

is that of the restarted problem with its own quadrature. Both predictions hold (Figure 3). The four restarted problems converge from the prior in 37–42 iterations to the restriction of the full optimum, within TV $= 3.5$–$5.5 \cdot 10^{-4}$, the level of the effective tolerance $\mathrm{tol}/\omega = 5 \cdot 10^{-4}$. Their minimizers are stable, with $\lambda_{\min}(t_{k_1}) = 0.0434$, 0.0433, 0.0433 and 0.0433 for $t_{k_1} = 0.2$, 0.4, 0.6 and 0.8, against 0.0442 for the full problem: restarting lowers $\lambda_{\min}$ by 2%, after which it hardly depends on t_{k_1}.

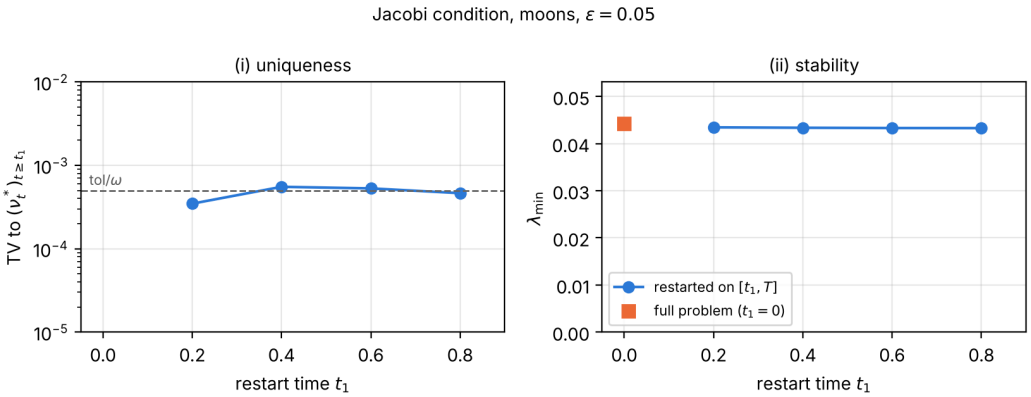


FIGURE 3. Jacobi condition (*moons*, $\varepsilon = 0.05$, pilot grid). Left: TV distance (max over layers) between the minimizer of the problem restarted at $(t_1, \gamma^*_{t_1})$ and the restriction of the full optimum; dashed: effective tolerance. Right: $\lambda_{\min}$ of the restarted problem, and of the full problem at $t_1 = 0$.

6. The PL inequality and the conjectured exponential convergence (Meta-Theorem 1.4)

6.1. **E1: the empirical PL constant $c(\varepsilon)$.** For each ε of the schedule we draw a cloud of 100 perturbations of $\boldsymbol{\nu}^*_\varepsilon$ on the asymmetric grid (mixture and multiplicative-tilt families, amplitudes $\delta \in \{0.005, 0.01, 0.02, 0.05, 0.1\}$, 10 per cell), evaluate $(J - J^*, \mathcal{I}_h)$ for each, and set $c(\varepsilon) = \inf \mathcal{I}_h/(J - J^*)$ over the admissible cloud ($\mathcal{I}_h \geq 10\,\mathcal{I}_h(\boldsymbol{\nu}^*_\varepsilon)$). On *moons* (Table 6): every point of every cloud is admissible at $\varepsilon \geq 0.05$ (100/100), every perturbation has $J > J^*$, all ratios are positive — the PL inequality holds throughout the sampled neighborhood — and the envelope follows a clean power law

$$c(\varepsilon) \;\propto\; \varepsilon^{\,1.06} \qquad \text{(log-log slope over } \varepsilon \in [0.02, 0.5];\ 0.90 \text{ over } [0.05, 0.5]). \tag{6.1}$$

At $\varepsilon = 0.02$ the admissibility filter rejects the entire cloud around the non-converged iterate ($\mathcal{I}_h = 8.9 \cdot 10^{-3}$ dominates); against the flow-refined endpoint (Section 6.2(c), $\mathcal{I}_h = 9 \cdot 10^{-11}$) instead, 99/100 points are admissible and $c(0.02) = 0.159$, extending the measured range to 1.4 decades, with local slopes $0.95, 0.86, 0.86, 1.62$: essentially linear, steepening at the smallest ε. The clouds probe the neighborhood in the exact sense of (2.4): the integrated relative entropy λ^2 ranges over $[1.2 \cdot 10^{-5}, 0.58]$ across the admissible clouds, so $c(\varepsilon)$ estimates the constant of Theorem 5.1 for entropy-radii up to $r^2 \approx 0.6$ at these data. Being an infimum over *sampled* directions, $c(\varepsilon)$ is an upper bound for the true local constant. On *housing* the picture is the same and cleaner. Every cloud is fully admissible (100/100, also at $\varepsilon = 0.02$ against the flow-refined endpoint, where $\mathcal{I}_h(\boldsymbol{\nu}^*) = 1.2 \cdot 10^{-17}$), every perturbation has $J > J^*$, and $c(\varepsilon) = 5.46$, 2.19, 1.10 and 0.227 at $\varepsilon = 0.5$, 0.2, 0.1 and 0.02 lie on a straight line of slope 0.99, with local slopes between 0.98 and 1.00: the degeneration is linear, with no steepening at small ε (Figure 4). [pending: *housing* cloud at $\varepsilon = 0.05$.]

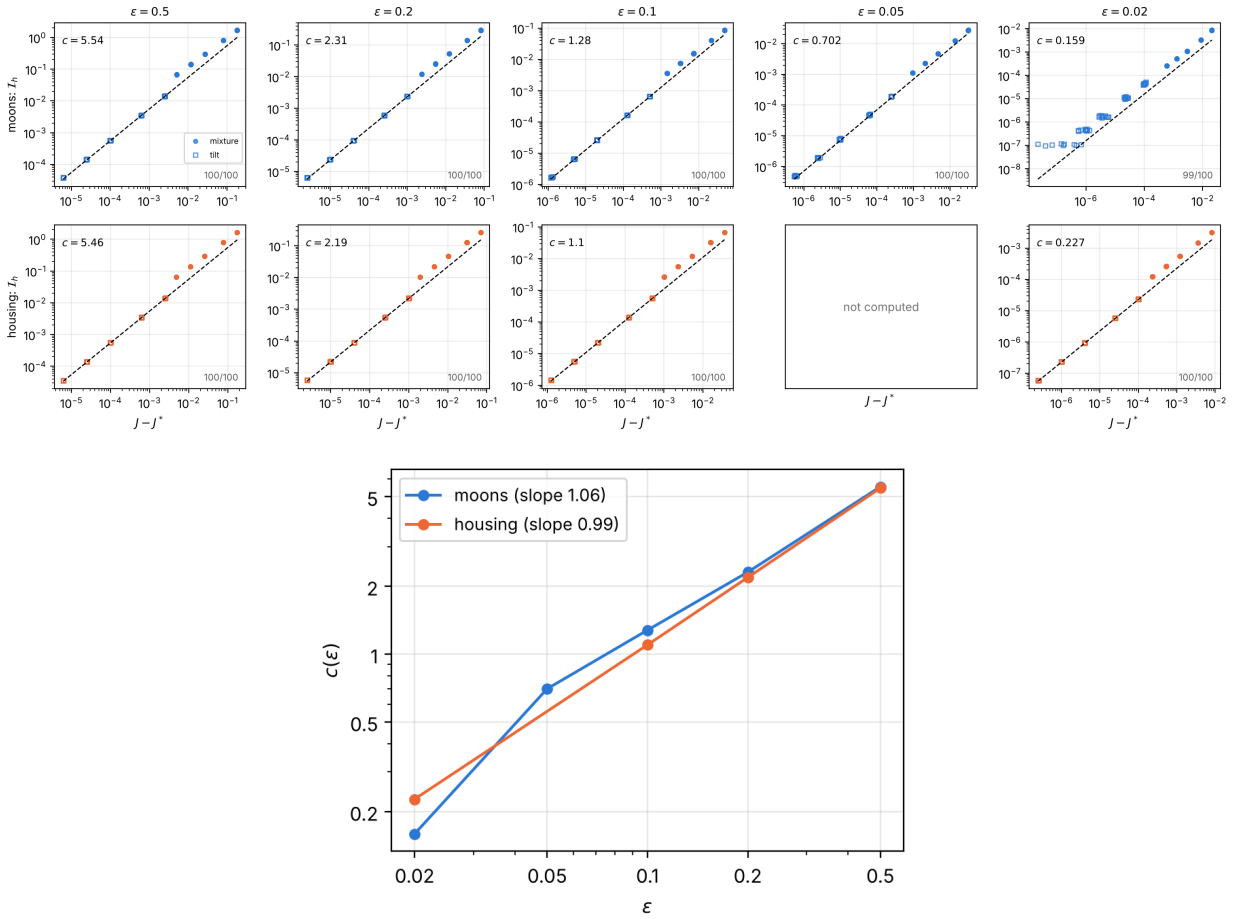


FIGURE 4. E1 (asymmetric grid). Top: $\mathcal{I}_h$ vs $J - J^*$ for the 100-point perturbation clouds at each ε (filled: mixtures, open: tilts; bottom right: admissible points), with the envelope $c(\varepsilon) = \inf \mathcal{I}_h/(J - J^*)$ (dashed). At $\varepsilon = 0.02$ the reference is the flow-refined endpoint. Bottom: $c(\varepsilon)$ in log-log with the least-squares slopes.

6.2. **E2: exponential convergence of the descent.** Descents are initialized at controlled TV distances from $\boldsymbol{\nu}^*$ ($\delta \in \{0.02, 0.05, 0.1\}$, seeds $0, 1, 2$; 600 steps, Ψ refresh $m = 5$, pilot grid) and the slope of $\log(J(\boldsymbol{\nu}^s) - J^*)$ on the decaying stretch (between $30\times$ and $1/3$ of the initial gap above the tolerance floor) gives $\hat{c}(\varepsilon)$. Three observations on *moons*:

(a) The conjectured exponential convergence holds. Every descent at $\varepsilon \geq 0.05$ decays exponentially over ≥ 4 decades before flooring at the solver-tolerance level $\sim (\mathrm{diffTV}/\omega)^2$ — e.g. at $\varepsilon = 0.05$ the gap falls from $1.6 \cdot 10^{-2}$ to $4 \cdot 10^{-7}$ —, and the floor itself confirms that the flow equilibrates at $\boldsymbol{\nu}^*$ within the solver tolerance. The decay is at least exponential but not a single exponential: within every fit window the first half decays 1.3–1.8 times faster than the second, as expected when the faster directions die out first and the slowest one takes over. The local rate stays bounded away from zero down to the floor, which is what the PL inequality asserts.

(b) The rate is a property of the minimizer, not of the initial direction. At fixed amplitude the three seeds give the same rate to better than 1% in 10 of the 15 (ε, δ) cells, and to 11% in the worst (e.g. $\varepsilon = 0.1$, $\delta = 0.1$: $\hat{c} = 0.5994, 0.5996, 0.6009$); this insensitivity to the initial direction is the PL phenomenon. The amplitude does matter: at every ε the rate at $\delta = 0.02$ exceeds the rate at $\delta = 0.1$ by a factor 1.2–1.8. This is the transient of (a), not a distance effect: a descent

started farther away spends more of its fit window in the regime governed by the slowest direction, so its fitted rate is lower. A PL constant that weakens away from $\boldsymbol{\nu}^*$ would instead make the decay accelerate as the gap closes, the opposite of what we see. The medians follow

$$\hat{c}(\varepsilon) \;\propto\; \varepsilon^{1.20} \quad (\varepsilon \in [0.02, 0.5]), \qquad \hat{c}/c \in [0.47, 0.74], \tag{6.2}$$

and the exponent hardly depends on the amplitude (1.17, 1.20 and 1.26 for $\delta = 0.02$, 0.05 and 0.1 separately). The ratio $\hat{c}/c$ decreases from 0.74 at $\varepsilon = 0.5$ to 0.47 at $\varepsilon = 0.05$ (0.50 at $\varepsilon = 0.02$). That it is below one is expected: the cloud envelope of E1 over-estimates the true constant, whereas the descent automatically finds the slowest direction. At the recovered endpoint $\varepsilon = 0.02$ the amplitude factor ($\hat{c} = 0.095$ at $\delta = 0.02$ vs 0.063 at $\delta = 0.1$, i.e. 1.5) is within the range seen at larger ε, so E2 gives no sign that the PL neighborhood shrinks with ε; E4 addresses that question directly.

(c) The non-converged endpoint undershoots — and the flow repairs it. At $\varepsilon = 0.02$ every descent drives J *below* $J(\boldsymbol{\nu}_{\mathrm{Picard}})$ immediately: with diffTV$/\omega = 0.55$ the last iterate of the fixed-point iteration is not the minimizer. Turning the observation into a tool, we integrate the flow *from* that iterate until $\mathcal{I}_h$ plateaus (the *flow-refined endpoint*). On the pilot grid the refinement moves the control by TV $= 0.55$ — exactly the calibrated defect —, lowers the cost from 0.3421 to 0.3192, drives $\mathcal{I}_h$ from $9 \cdot 10^{-3}$ to 10^{-8}, and its fixed-point residual after one *undamped* Gibbs update is $\mathrm{TV}(\nu_{\mathrm{Gibbs}}, \nu) = 2.6 \cdot 10^{-7}$: the refined control *is* the Gibbs fixed point, reached where the fixed-point iteration has lost contractivity (Section 3.1). Test accuracy is essentially unchanged (0.930 vs 0.935). This recovers the $\varepsilon = 0.02$ endpoint for E1/E2 and suggests *Picard-then-flow* as the standard pipeline at small ε. Figure 5 shows the decay curves.

(d) Convergence of the trajectory. Beyond the cost, the authors suggest that the control itself should converge exponentially (cf. [9]). Along the same descents we record the integrated relative entropy $\int \mathcal{E}(\nu_t^s | \nu_t^*)\, dt$, the very distance in which Theorem 5.1 measures the PL neighborhood. At $\varepsilon \in \{0.5, 0.2\}$, where it was recorded on both benchmarks, it decays exponentially at the same rate as the cost: the median fitted rates are 4.13 and 1.54 on *moons* (against $\hat{c} = 4.12$ and 1.41) and 3.97 and 1.61 on *housing* (against 3.98 and 1.55), ratios between 1.00 and 1.09. Since $J - J^*$ and the integrated entropy are both quadratic in $\boldsymbol{\nu} - \boldsymbol{\nu}^*$ near the minimizer, equal rates are what exponential convergence of the control itself predicts. [pending: the entropy decay at $\varepsilon \le 0.1$.]

On *housing* the descents at $\varepsilon = 0.5$ and 0.2 behave as on *moons*: median rates $\hat{c} = 3.98$ and 1.55, seeds agreeing to better than 0.1% in 5 of the 6 (ε, δ) cells, and the same amplitude factor (1.26 and 1.19). [pending: *housing* descents at $\varepsilon \in \{0.1, 0.05\}$ and at the flow-refined $\varepsilon = 0.02$ endpoint.]

6.3. **E4: radius of the PL neighborhood.** We sweep mixture perturbations of increasing amplitude $\delta \in [0.02, 0.9]$ (6 per amplitude, *moons*, $\varepsilon = 0.05$, asymmetric grid) and monitor the worst ratio $\mathcal{I}_h/(J - J^*)$ against the small-amplitude envelope $c(0.05) = 0.70$ of E1. No boundary is visible: the minimal ratio stays in [0.89, 1.08] across the whole sweep — it dips mildly around $\delta \approx 0.1$–0.2 and *increases* again for $\delta \ge 0.5$ — and not a single perturbation falls below $c(0.05)$. Within this perturbation family the local PL inequality is, in practice, global at moderate ε: the entropy radius r exceeds everything we can reach with TV ≤ 0.9 (λ^2 up to 7.2). Nor does E2 show a finite neighborhood: the amplitude dependence of $\hat{c}$ is present at every ε, is no stronger at $\varepsilon = 0.02$, and goes the wrong way for one (Section 6.2(b)). Whether the radius shrinks with ε is therefore open. The caveat is plain: one perturbation family, one ε.

6.4. **The degeneration picture.** The two completed estimators on *moons* give power laws with exponents close to one: the PL constant of [3] degenerates *essentially linearly in* ε, which quantifies the non-uniformity in ε that the authors flag as an open direction. Figure 6 superposes the three estimators in log-log. $\lambda_{\min}(\varepsilon)$ — which lives in a different (Euclidean, grid-dependent) normalization and sits an order of magnitude below — degenerates faster on *moons* (slope 1.43) and shares with c and $\hat{c}$ the super-linear steepening at the smallest ε. On *housing* the degeneration is linear without steepening: $c \propto \varepsilon^{0.99}$ down to $\varepsilon = 0.02$ and $\lambda_{\min} \propto \varepsilon^{0.90}$.

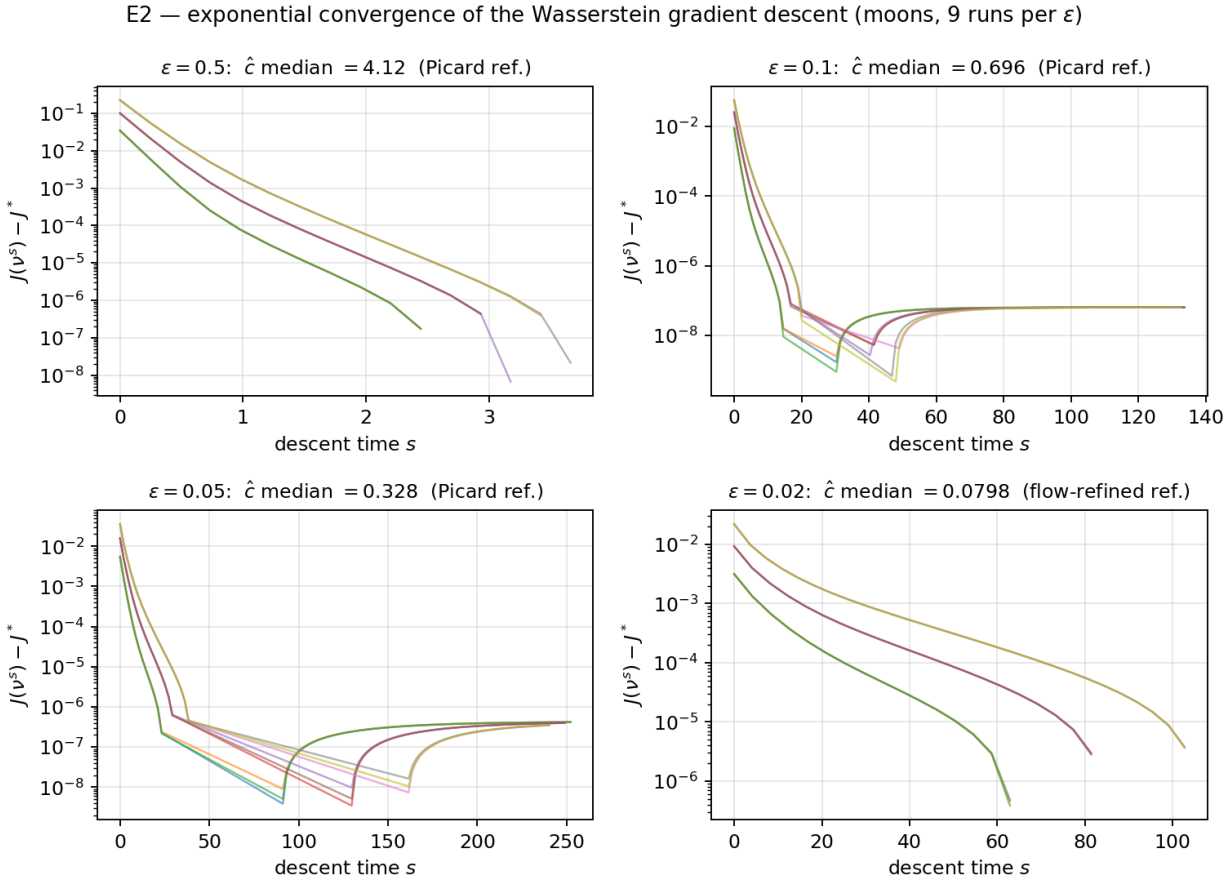


FIGURE 5. E2 (*moons*, pilot grid): $J(\boldsymbol{\nu}^s) - J^*$ for the nine (amplitude, seed) descents at four values of ε. The decay is exponential over ≥ 4 decades down to the solver-tolerance floor; at $\varepsilon = 0.02$ the reference is the flow-refined endpoint.

	moons			*housing*		
ε	c (E1)	$\hat{c}$ (E2)	$\lambda_{\min}$ (E3)	c (E1)	$\hat{c}$ (E2)	$\lambda_{\min}$ (E3)
0.5	5.54	4.12	1.233	5.46	3.98	1.062
0.2	2.31	1.41	0.368	2.19	1.55	pend.
0.1	1.28	0.70	0.144	1.10	pend.	0.236
0.05	0.70	0.33	0.0442	pend.	pend.	0.134
0.02†	0.159	0.080	pend.	0.227	pend.	pend.
log-log slope	1.06	1.20	1.43	0.99	pend.	0.90

TABLE 6. The three estimators of the PL degeneration (E1 on the asymmetric grid; E2, E3 on the pilot grid; $\hat{c}$ is the median of 9 descents). Slopes over the computed values. †Relative to the flow-refined endpoint (99/100 admissible cloud points on *moons*, 100/100 on *housing*; the non-converged iterate admits none on *moons*).

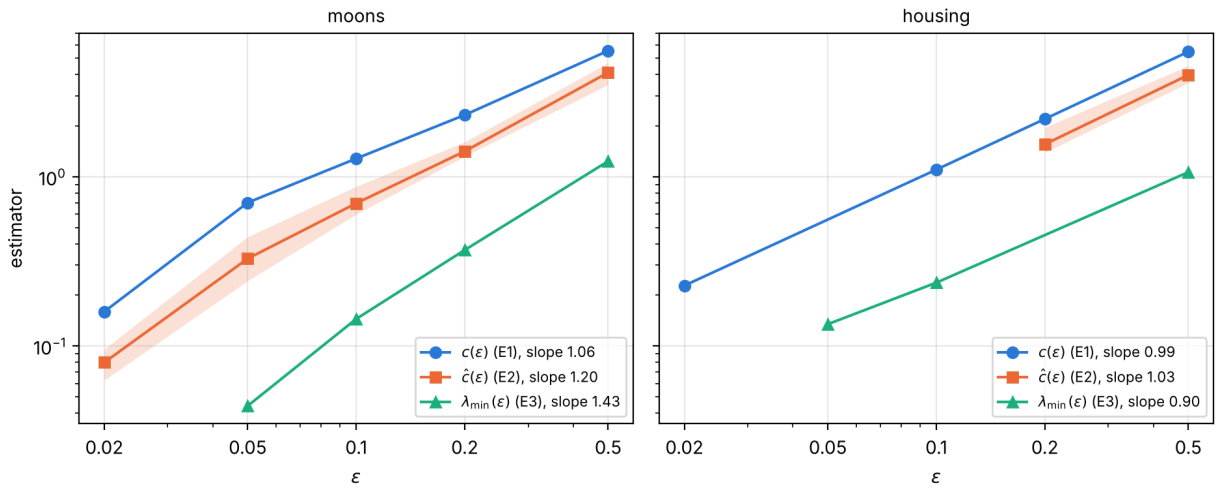


FIGURE 6. The three independent estimators of the PL degeneration in log-log, with least-squares slopes; the band is the min–max of the nine descents of E2. Left: *moons*. Right: *housing*.

7. Discussion

7.1. **Why (essentially) linear degeneration: a heuristic.** The measured exponents cluster around 1, and a two-line expansion suggests why this should be the leading behavior. Write $\nu_t = \nu_t^*(1+\varphi_t)$ for a small relative perturbation and let $\eta = \nu - \boldsymbol{\nu}^*$. Since the Gibbs formula cancels the integrand of $\mathcal{I}$ at $\boldsymbol{\nu}^*$,

$$\mathcal{I}(\boldsymbol{\nu}) \;\approx\; \int_{t_0}^{T}\!\!\int_A \big|\,\varepsilon\nabla_a\varphi_t + \delta[\nabla_a\Psi_t](\eta)\,\big|^2\, d\nu_t^*\, dt, \qquad J - J^* \;\approx\; \tfrac{1}{2}\Big[\varepsilon\int_{t_0}^{T}\!\!\int_A \varphi_t^2\, d\nu_t^*\, dt + \langle H_\Psi\eta,\eta\rangle\Big],$$

where $\delta[\nabla_a\Psi](\eta)$ and H_Ψ carry the terminal cost and are $O(\eta)$ uniformly in ε, while the entropic contributions carry explicit factors ε^2 (in $\mathcal{I}$) and ε (in $J - J^*$). For directions nearly invisible to the terminal cost — abundant, since the data see the parameter space only through N particles — the ratio reduces to a Rayleigh quotient,

$$\frac{\mathcal{I}}{J - J^*} \;\approx\; 2\varepsilon\,\frac{\iint |\nabla_a\varphi|^2\, d\nu^*\, dt}{\iint \varphi^2\, d\nu^*\, dt} \;\geq\; 2\varepsilon\,\rho(\boldsymbol{\nu}_\varepsilon^*),$$

with ρ the (worst-slice) Poincaré constant of ν_ε^*. If $\rho(\boldsymbol{\nu}_\varepsilon^*) = O(1)$ over the measured range, $c(\varepsilon) \propto \varepsilon$. The deviations from slope one then encode the ε-dependence of the spectral constants of ν_ε^* itself, i.e. how the concentration of the Gibbs measure (Table 3) competes with the vanishing prefactor. This is consistent with the steepening being visible on *moons*, where ν_ε^* concentrates markedly as ε decreases, and with the behavior on *housing*, where it does not: there the local slopes of $c(\varepsilon)$ stay between 0.98 and 1.00 down to $\varepsilon = 0.02$, as predicted for $\rho(\boldsymbol{\nu}_\varepsilon^*) = O(1)$. The same expansion predicts $\lambda_{\min}(\varepsilon) \propto \varepsilon$ for the entropic block of the Hessian, up to the grid-dependent Euclidean normalization; only the *slope* of $\lambda_{\min}(\varepsilon)$ is comparable across estimators. In the terminology of [3], the local PL constant is not merely non-uniform but vanishes at an essentially linear rate, consistent with the entropic term being the sole source of local convexity; in the unregularized limit the landscape changes nature altogether [7].

7.2. **Methodological recommendations.** (i) Report diffTV/ω, not diffTV: it is the distance all downstream tolerances inherit. (ii) When a damped fixed-point iteration stalls, check for a period-2

cycle before blaming the continuation step. (iii) Discretize first variations compatibly, so that the discrete minimizer is an exact critical point. (iv) Use exponential-fitting fluxes for measure-space gradient flows, so that the discrete equilibrium is the discrete Gibbs measure; the flow is then also a robust solver where the fixed-point iteration fails. (v) Split Hessian-vector products into the exact entropic diagonal plus differences of the smooth part, and remove the projected null space before computing $\lambda_{\min}$.

7.3. **Limitations and outlook.** The solver meshes A, so everything lives in low dimension ($d' = 5$); generalizing beyond tensor grids (adaptive grids concentrated on the effective support of Table 3, sparse grids, particle discretizations) is open. The clean measurements span 1.4 decades of ε, bounded below by the loss of contractivity of the fixed-point iteration; Newton-type solvers or ε-dependent damping would extend the range, subject to grid resolution. Our genericity probe samples data sets near two benchmarks and cannot exhibit the complement of $\mathcal{O}$; locating non-generic data (e.g. symmetric data sets where several minimizers could coexist) is a natural next step. Finally, $\hat{c}$ is measured through one discretization of the Wasserstein flow; its agreement in slope with the discretization-independent ratio estimator $c(\varepsilon)$ is our main evidence that the measured degeneration is not a numerical artifact. Two questions raised in [3] remain for future work: the convergence of J^* and $\boldsymbol{\nu}^*$ as the number N of data points grows (with rates expected to improve when $\gamma_0 \in \mathcal{O}$), and the approximation of $\boldsymbol{\nu}^*$ by networks with finitely many neurons.

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Appendix A. Characteristics and the Jacobian cocycle

With the mean field frozen, the characteristics of the backward transport equation $-\partial_t u_t - \bar{b}(x, \nu_t) \cdot \nabla_x u_t = 0$, $u_T = L$, are the forward particle trajectories $\dot{X}_t = \bar{b}(X_t, \nu_t)$, and $\frac{d}{dt} u_t(X_t, y) = \partial_t u_t + \nabla_x u_t \cdot \bar{b} = 0$. Hence $u_t(X_t(x_0), y) = L(X_T(x_0), y)$ for every x_0. Differentiating in x_0 with $\Phi_s = \partial X_s / \partial x_0$ gives $\Phi_t^\top \nabla_x u_t(X_t, y) = \Phi_T^\top \nabla_x L(X_T, y)$, and since the flow is a diffeomorphism, $\nabla_x u_t(X_t, y) = [\Phi_T \Phi_t^{-1}]^\top \nabla_x L(X_T, y)$ — formula (3.1). The Jacobians solve $\dot{\Phi} = \partial_x \bar{b}(X, \nu_t)\Phi$, $\Phi(0) = I$, and are integrated by RK4 jointly with the particles. Meshing $\mathbb{R}^{d_1}$ for the backward PDE would scale exponentially in d_1; the cocycle costs $O(N d_1^3)$ per slice and needs no spatial mesh.

Appendix B. The exact $a \leftrightarrow -a$ symmetry

Proposition B.1 ($a \leftrightarrow -a$ invariance of the optimal control)**.** *With $\sigma = \tanh$ (odd) and ℓ even, any control given by the Gibbs formula* (2.3) *satisfies, for every t,*

$$\nu_t^*(-a) = \nu_t^*(a) \quad \forall a \in A, \qquad \textit{hence} \qquad \mathbb{E}_{\nu_t^*}[a] = 0.$$

The invariance is structural: it depends only on the parities of $b(x, \cdot)$ and ℓ, not on γ_0, L or ε.

Proof. Write the Gibbs exponent as $-\ell(a) - \Psi_t(a)/\varepsilon$ with Ψ_t as in (2.3). (i) $\ell(-a) = \ell(a)$ since ℓ depends on $|a|$. (ii) Negating the three blocks of $a = (a_0, a_1, a_2)$ and using that tanh is odd,

$$b(x, -a) = \tanh(-a_1 \cdot x - a_2)(-a_0) = \tanh(a_1 \cdot x + a_2)\, a_0 = b(x, a),$$

so $b(x, \cdot)$ is *even*. (iii) The pair (γ_t, u_t) does not depend on a, hence $\Psi_t(-a) = \Psi_t(a)$ by (ii). The exponent is therefore even, giving $\nu_t^*(-a) = \nu_t^*(a)$; and $a\, \nu_t^*(a)$ is odd, so its integral over the symmetric domain vanishes. □

Remark B.2. The zero first moment reflects the mirror redundancy of weight space ($b(x,-a) = b(x,a)$): a and $-a$ implement the same map and their contributions to $\bar{b}$ reinforce), not a cancellation of the control. On our odd-m_j grids, which are symmetric under $a \mapsto -a$, the Gibbs update produces exactly even node weights, so the discrete ν^* is symmetric to machine precision (error 0.0 bitwise along the whole schedule, $\max_i |\mathbb{E}[a_i]| \lesssim 6 \cdot 10^{-16}$), and the discrete flow preserves it. Any asymmetric bug in the forward pass, the cocycle or the Gibbs update breaks this test immediately.

Appendix C. Reproducibility

All code, frozen $\boldsymbol{\nu}^*$ checkpoints and experiment outputs are available at `https://github.com/dantor03/mfnode-pl-verification` [TODO: tag the submission commit]. The library is pure Python (≥ 3.10) on `numpy`/`scipy`/`scikit-learn`; no GPU is used. The correctness suite (`pytest tests/`, 18 tests, ≈ 4 s) covers every structural property quoted in Section 3.5.

Data. Every random object derives from the single seed SEED = 42. *moons*: `make_moons` with $n = 400$, noise 0.12, standardized; labels embedded as anchors $(\pm 1, 0)$. *housing*: California Housing, subsample of 600 rows (400 train, 200 test, split seed 42), features and target standardized, restricted to the two features with largest |corr| with the target on the train split (MedInc, AveOccup). Genericity probe (Section 5.3): *moons* with sampling seeds 0–4 at noise 0.12, and seed 42 at noise 0.08 and 0.16; *housing* with subsample and split seeds 0–4 (600 rows, 400 train), the two features fixed to MedInc and AveOccup so that only the sample changes.

Hyperparameters.

horizon / slices	$T = 1$, $K = 10$ (trapezoidal w_k)
prior	$\ell(a) = 0.05\|a\|^4 + 0.5\|a\|^2$
parameter box	$A = [-2, 2]^5$
grids	pilot $m_j \equiv 5$ ($M = 3125$); asymmetric $m = (7, 9, 5, 7, 5)$ ($M = 11025$)
fixed-point iteration	$\omega = 0.2$, tol $= 10^{-4}$ on diffTV, max_iter $= 200$
continuation	$\varepsilon \in (0.5, 0.2, 0.1, 0.05, 0.02)$, warm starts; fine: $(0.04, 0.03, 0.025, 0.02)$
Fisher	compatible (fd) assembly, $\log \nu$ floored at -40
flow	Scharfetter–Gummel, explicit Euler at $0.9\times$CFL, Ψ refreshed every $m = 5$ steps, per-slice renormalization
Hessian	central step $h = 10^{-5}$, support threshold 10^{-6}, LOBPCG with entropic-diagonal preconditioner, null-space penalty $\sigma = 100$

Timings. Single laptop CPU: ≈ 9 s (pilot) and ≈ 35 s (asymmetric) per fixed-point iteration; full continuations 0.5–5 h; E1 ≈ 2 h and E2 1–2 h per ε; E3 0.3–3.6 h per eigenvalue; flow refinement of the $\varepsilon = 0.02$ endpoint 1 h (pilot) to 7 h (asymmetric). Every $\boldsymbol{\nu}^*$ is stored as a resumable checkpoint consumed unchanged by all experiment runners.

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